

the “domain” of the object or the event. The classroom is a domain; real life is another. We react to a piece of information not on its logical merit, but on the basis of which framework surrounds it, and how it registers with our social-emotional system. Logical problems approached one way in the classroom might be treated differently in daily life. Indeed they *are* treated differently in daily life.

Knowledge, even when it is exact, does not often lead to appropriate actions because we tend to forget what we know, or forget how to process it properly if we do not pay attention, even when we are experts. Statisticians, it has been shown, tend to leave their brains in the classroom and engage in the most trivial inferential errors once they are let out on the streets. In 1971, the psychologists Danny Kahneman and Amos Tversky plied professors of statistics with statistical questions not phrased as statistical questions. One was similar to the following (changing the example for clarity): Assume that you live in a town with two hospitals—one large, the other small. On a given day 60 percent of those born in one of the two hospitals are boys. Which hospital is it likely to be? Many statisticians made the equivalent of the mistake (during a casual conversation) of choosing the larger hospital, when in fact the very basis of statistics is that large samples are more stable and should fluctuate less from the long-term average—here, 50 percent for each of the sexes—than smaller samples. These statisticians would have flunked their own exams. During my days as a quant I counted hundreds of such severe inferential mistakes made by statisticians who forgot that they were statisticians.

For another illustration of the way we can be ludicrously domain-specific in daily life, go to the luxury Reebok Sports Club in New York City, and look at the number of people who, after riding the escalator for a couple of floors, head directly to the StairMasters.

This domain specificity of our inferences and reactions works both ways: some problems we can understand in their applications but not in textbooks; others we are better at capturing in the textbook than in the practical application. People can manage to effortlessly solve a problem in a social situation but struggle when it is presented as an abstract logical problem. We tend to use different mental machinery—so-called modules—in different situations: our brain lacks a central all-purpose computer that starts with logical rules and applies them equally to all possible situations.

And as I’ve said, we can commit *a logical mistake in reality but not in the classroom*. This asymmetry is best visible in cancer detection. Take doctors examining a patient for signs of cancer; tests are typically done on patients who want to know if they are cured or if there is “recurrence.” (In fact, recurrence is a misnomer; it simply means that the treatment did not kill all